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LAB 03

Report

Overview

This laboratory session provided practical exposure to using a professional multi-function network cable tester. The purpose was to evaluate the continuity, wiring arrangement, and termination quality of different cable types such as RJ-45 (Ethernet), RJ-11 (telephone), USB, and BNC (coaxial). By analyzing LED indicators on the tester, various faults were identified, including short circuits, open connections, crossed wires, and improper terminations. The activity emphasized the importance of cable testing in maintaining reliable communication systems and preventing network failures.

1. Introduction

In computer networking, the physical layer is essential because it supports all higher-level communication processes. Even with advanced hardware and protocols, faulty cabling can disrupt performance. Issues such as incorrect wiring, damaged cables, or poor crimping may result in slow speeds, data loss, or complete disconnection.

This lab session focused on developing practical skills using a handheld cable tester. Learning to diagnose cable issues efficiently is crucial for professionals working in network installation, troubleshooting, and maintenance.

2. Objectives

The main goals of this experiment were:

- To understand how a multi-function cable tester operates.
- To test different cable types (RJ-45, RJ-11, USB, and BNC) for proper connectivity.
- To detect common faults such as shorts, open wires, and miswiring.

- To recognize the role of cable testing in ensuring network reliability and compliance with standards like TIA/EIA-568.

3. Theory and Background

Network cables act as transmission media for signals between devices. Common types include Ethernet cables (RJ-45), telephone cables (RJ-11), USB cables, and coaxial cables (BNC).

Cable testers function by sending signals through each conductor and analyzing the returned signal. This helps identify:

- Proper continuity between cable ends
- Shorts caused by unintended connections
- Open circuits due to broken wires
- Incorrect wiring configurations
- Termination issues in coaxial cables

The tester used in this lab included dedicated ports for multiple cable types and LED indicators to display test results and fault conditions.

4. Apparatus

- Multi-function network cable tester
- Assorted network cables (RJ-45, RJ-11, USB, BNC)
- Notebook for observations

5. Methodology

1. The tester was first examined to understand its controls and indicators.
2. The device was powered on, and battery status was checked.

3. Each cable type was connected to the appropriate port.
4. The tester automatically scanned each wire, and LED patterns were observed.
5. Both functional and faulty cables were tested to compare results.

6. Results and Observations

Most tested Ethernet cables showed correct operation, indicated by sequential LED lighting and a **CONNECTED** signal. This confirmed proper wiring and continuity.

During testing, the device displayed active pin checking through illuminated LEDs, and the battery indicator confirmed proper power status. Faulty cables (where tested) produced different LED patterns, helping identify specific issues.

Network Cable Tester in Operation

7. Discussion

The experiment highlighted how even small wiring faults can significantly affect network performance.

For example, a mis wired pair in an Ethernet cable can reduce transmission efficiency or cause errors.

The cable tester proved to be a simple yet effective tool for diagnosing such issues quickly. This aligns with theoretical knowledge of the physical layer and demonstrates real-world practices used in network maintenance.

8. Conclusion

The lab successfully demonstrated the use of a network cable tester and reinforced the importance of proper cabling in network systems. The ability to identify faults and verify cable integrity is essential for ensuring efficient and reliable communication.

Future work could include creating cables manually and testing them or using advanced testers capable of measuring cable length and signal quality.

*****THE END*****